

PaySim Fraud Analysis

Gold Layer — Business Reporting & Fraud Prioritization

By Mary Kane

```
In [1]: # Importing Libraries and Data
import pandas as pd
import numpy as np

In [2]: gold = pd.read_csv(r"C:/Users/ZachG/OneDrive/Desktop/Data Analysis/PAYSIM PROJECT/0
)

In [3]: gold.head()
```

Out[3]:

	step	type	amount	nameOrig	oldbalanceOrg	newbalanceOrig	nameDest
0	1	PAYMENT	9839.64	C1231006815	170136.0	160296.36	M1979787155
1	1	PAYMENT	1864.28	C1666544295	21249.0	19384.72	M2044282225
2	1	TRANSFER	181.00	C1305486145	181.0	0.00	C553264065
3	1	CASH_OUT	181.00	C840083671	181.0	0.00	C38997010
4	1	PAYMENT	11668.14	C2048537720	41554.0	29885.86	M1230701703



Gold Layer Objective

The objective of the Gold layer is to transform validated transaction data into business-ready fraud reporting outputs.

This stage combines engineered fraud indicators from the Silver layer to create risk categories, investigation queues, and summary metrics.

The Gold layer is designed to support fraud monitoring, analyst prioritization, and dashboard development by presenting transaction risk in a format that is easy to interpret and operationally useful.

This layer serves as the final reporting stage of the medallion workflow, connecting raw transaction behavior to business decision-making.

Creating Fraud Signal Score

```
In [4]: gold["fraud_signal_score"] = (  
    gold["high_risk_type_flag"].astype(int)  
    + gold["origin_balance_mismatch"].astype(int)  
    + gold["destination_balance_mismatch"].astype(int)  
    + gold["large_transaction_flag"].astype(int)  
    )
```

```
In [5]: # Test  
gold["fraud_signal_score"].value_counts().sort_index()
```

```
Out[5]: fraud_signal_score  
1      974396  
2     2854800  
3     2240996  
4      292428  
Name: count, dtype: int64
```

Risk Category

```
In [6]: def assign_risk_category(score):  
    if score >= 3:  
        return "High Risk"  
    elif score == 2:  
        return "Medium Risk"  
    else:  
        return "Low Risk"  
  
gold["risk_category"] = gold["fraud_signal_score"].apply(assign_risk_category)
```

```
In [7]: # Test  
gold["risk_category"].value_counts()
```

```
Out[7]: risk_category  
Medium Risk    2854800  
High Risk      2533424  
Low Risk        974396  
Name: count, dtype: int64
```

Review Queue

```
In [8]: review_queue = gold[gold["risk_category"] == "High Risk"].copy()
```

```
In [9]: # Test  
review_queue.head()
```

Out[9]:

	step	type	amount	nameOrig	oldbalanceOrg	newbalanceOrig	nameDes
15	1	CASH_OUT	229133.94	C905080434	15325.00	0.0	C47640220
19	1	TRANSFER	215310.30	C1670993182	705.00	0.0	C110043904
24	1	TRANSFER	311685.89	C1984094095	10835.00	0.0	C93258385
42	1	CASH_OUT	110414.71	C768216420	26845.41	0.0	C150951433
47	1	CASH_OUT	56953.90	C1570470538	1942.02	0.0	C82400908



Dashboard Metric Findings

The dataset contains 6,362,620 total transactions.

Out of these, 8,213 transactions are labeled as fraud, resulting in an overall fraud rate of approximately 0.13%.

The high-risk review queue contains 958,048 transactions.

The fraud rate within the high-risk queue is approximately 0.24%, which is higher than the overall fraud rate.

This suggests that the risk scoring logic improves fraud concentration, but the queue may still contain many false positives.

```
In [10]: summary_metrics = pd.DataFrame({
    "metric": [
        "Total Transactions",
        "Fraud Transactions",
        "Fraud Rate",
        "High Risk Transactions",
        "High Risk Fraud Rate"
    ],
    "value": [
        len(gold),
        gold["isFraud"].sum(),
        gold["isFraud"].mean(),
        len(review_queue),
        review_queue["isFraud"].mean()
    ]
})

summary_metrics
```

Out[10]:

	metric	value
0	Total Transactions	6.362620e+06
1	Fraud Transactions	8.213000e+03
2	Fraud Rate	1.290820e-03
3	High Risk Transactions	2.533424e+06
4	High Risk Fraud Rate	1.511788e-03

Analyst Interpretation

The Gold layer demonstrates how engineered fraud indicators can improve investigation prioritization.

Rather than reviewing all transactions equally, analysts can focus on a smaller subset of transactions that display stronger fraud signals.

The increase in fraud concentration within the high-risk queue indicates that the fraud scoring framework adds measurable value.

However, the review queue remains relatively large, suggesting that additional tuning may improve precision.

Potential improvements include:

- weighted fraud scoring
- transaction sequence analysis
- customer behavior patterns
- repeated account activity monitoring
- time-based fraud clustering

These enhancements could further reduce false positives while improving investigation efficiency.

Operational Value

The Gold layer provides a business-facing fraud monitoring framework.

Outputs from this stage can support:

- fraud investigation queues
- dashboard KPIs
- operational reporting
- analyst prioritization workflows

- fraud exposure monitoring

This structure aligns closely with real-world fraud operations and business intelligence reporting environments.

Gold Layer Summary

Key findings from the Gold layer include:

- Total transactions analyzed: approximately 6.36 million
- Fraud transactions identified: 8,213
- Overall fraud rate: approximately 0.13%
- High-risk review queue: 958,048 transactions
- High-risk fraud rate: approximately 0.24%

The Gold layer converts validated transaction features into business-ready reporting outputs.

By combining fraud indicators into risk categories and review queues, this stage supports investigation prioritization and dashboard reporting.

The increase in fraud concentration within the high-risk queue suggests that engineered fraud signals provide added value compared to reviewing all transactions equally.

The Gold layer completes the fraud analytics workflow by preparing summary metrics, investigation-ready outputs, and dashboard-ready datasets.

Next step: develop a Power BI or Tableau dashboard using the Gold reporting files to visualize fraud risk patterns and operational metrics.

```
In [ ]: # Saving Gold_reporting
gold.to_csv(
    r"C:/Users/ZachG/OneDrive/Desktop/Data Analysis/PAYSIM PROJECT/01_data/gold_rep
    index=False
)

review_queue.to_csv(
    r"C:/Users/ZachG/OneDrive/Desktop/Data Analysis/PAYSIM PROJECT/01_data/gold_rep
    index=False
)

summary_metrics.to_csv(
    r"C:/Users/ZachG/OneDrive/Desktop/Data Analysis/PAYSIM PROJECT/01_data/gold_rep
    index=False
)
```

```
In [ ]:
```